Carrier board v1: soldering guide

Board 154 x 96 mm, bare copper (no printing). Every part goes on the TOP; every pin is soldered on the BOTTOM. Page 2: what the parts look like. Pages 3-10: step by step, one part per page. Pages 11-13: whole board and the unplated-hole details. Page 14: checks.

Which side is which

1. Lay the board so the row of 4 large 5-pin footprints (module plugs J1-J4) is along the edge nearest you.
2. TOP (parts side): the long double row of 24 pads (Teensy socket) is at the LEFT, starting at the left edge; the short double row of 6 (DAC socket) is right of it, near the far edge. The 2 left mounting holes are lower (32 and 58 mm from the far edge) than the 2 right ones (10 and 45 mm).
3. If the Teensy rows are at the right instead, you are looking at the BOTTOM (solder side).
4. Check against page 11: the tracks you see on the top must match the red tracks there.

Before you start

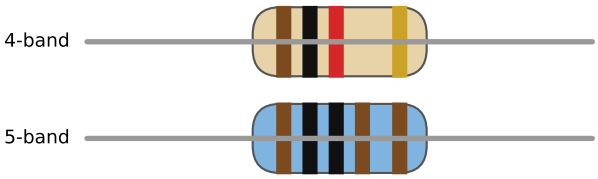
1. Look into a few holes. Copper-lined (shiny ring inside) = plated: solder on the bottom only. Bare fibreglass inside = NOT plated (this board): the steps already include it; details on page 13.
2. The board takes FEMALE headers (sockets). The Teensy and the DAC keep their male pins and plug in from the top. Never solder male pins into the board, and never solder the Teensy or DAC to it.
3. Multimeter in beep mode at hand. No Teensy, no DAC, no modules on the board while soldering.

Parts and order (lowest first; what they look like: next page)

#	Where	Part	How
1	R1-R4	4 x 1 kOhm resistor	either way round; holes 10 mm apart
2	R5-R8	4 x 1 MOhm resistor	either way round; holes 7.6 mm apart
3	D1-D4	4 x BAT85 diode (glass)	BAND towards the far edge (Teensy side): step 3
4	U1	2 x female header 1x24, 8.5 mm	Teensy socket, two rows
5	U2	2 x female header 1x6	DAC socket, two rows
6	J1-J4	4 x Phoenix MSTB 2,5/5-GF-5,08	plug opening towards the board edge

What the parts look like

Drawings, not to the same scale; real sizes in the text. German drawer labels in quotes. Bring this page and a multimeter: the resistor colours can be confusing, the meter is sure (1 kOhm reads 0.99-1.01 k, 1 MOhm 0.99-1.01 M).



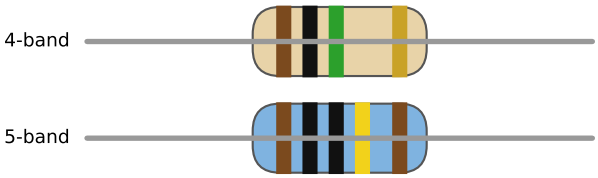
R1-R4: resistor 1 kOhm (x4)

"Widerstand 1 k" (1 kOhm, 1K0). Small cylinder, 6 mm long, 2.5 mm thick, a wire at each end.

4 bands (beige body): brown-black-RED-gold.

5 bands (blue body, 1 %): brown-black-black-BROWN-brown.

0.25 W (any size this small is fine).



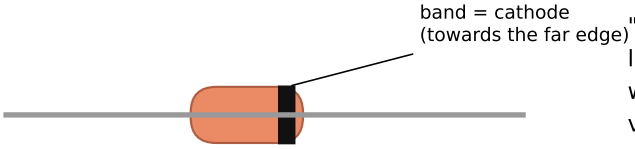
R5-R8: resistor 1 MOhm (x4)

"Widerstand 1 M" (1 MOhm, 1M0). Same size as the 1 k: only the colours differ.

4 bands: brown-black-GREEN-gold.

5 bands: brown-black-black-YELLOW-brown.

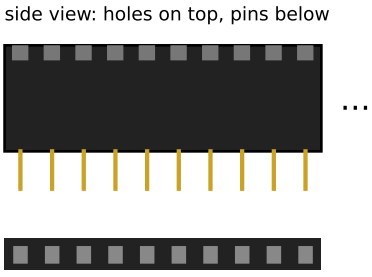
Do not mix up with 1 k: the 3rd (4-band) or 4th (5-band) colour is the difference.



D1-D4: diode BAT85 (x4)

"Schottky-Diode BAT85". Tiny glass body (3.8 mm long, 1.9 mm thick), orange-red and see-through, with a black band at one end = cathode. Printed text very small: use a magnifier.

Substitutes if there is no BAT85: BAT43, BAT46, BAT42 (same job). NOT a 1N4148 (that is the wrong type).

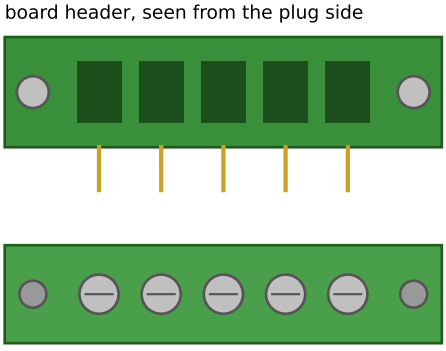


U1, U2: female headers (sockets)

"Buchsenleiste 2,54 mm, 1-reihig". Black plastic strip with a row of square holes on top, pins underneath. Need 2 x 24-pin (Teensy, 8.5 mm tall) and 2 x 6-pin (DAC).

Longer strips (1 x 40) are fine: cut them with side cutters THROUGH the next hole (you lose one pin), file flat.

NOT the "Stiftleiste" (male pin header: bare pins sticking up). Spare male strips only help hold the sockets straight while soldering.



J1-J4: Phoenix MSTB header (x4) + plug (x4)

On the board: "Leiterplatten-Grundleiste" Phoenix MSTB 2,5/5-GF-5,08 (no. 1776537): green, 5 pin openings 5.08 mm apart, a threaded hole in a flange at each end, 5 pins underneath. The opening faces sideways (horizontal).

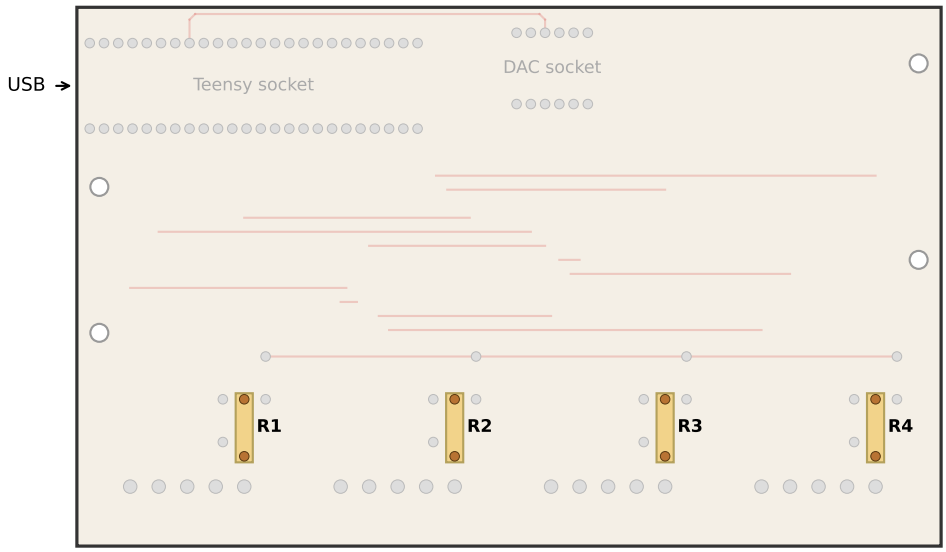
On each module cable: the plug MSTB 2,5/5-STF-5,08 (1778014): green, 5 screw terminals on top, wires in the side, a screw in each flange. Same family = they fit; other pitches (3.5 / 3.81 mm) do not.

No Phoenix in the makerspace? Order them (README): no substitute keeps the locking flanges.

Step 1: R1-R4, 1 kOhm (4 resistors, 8 joints)

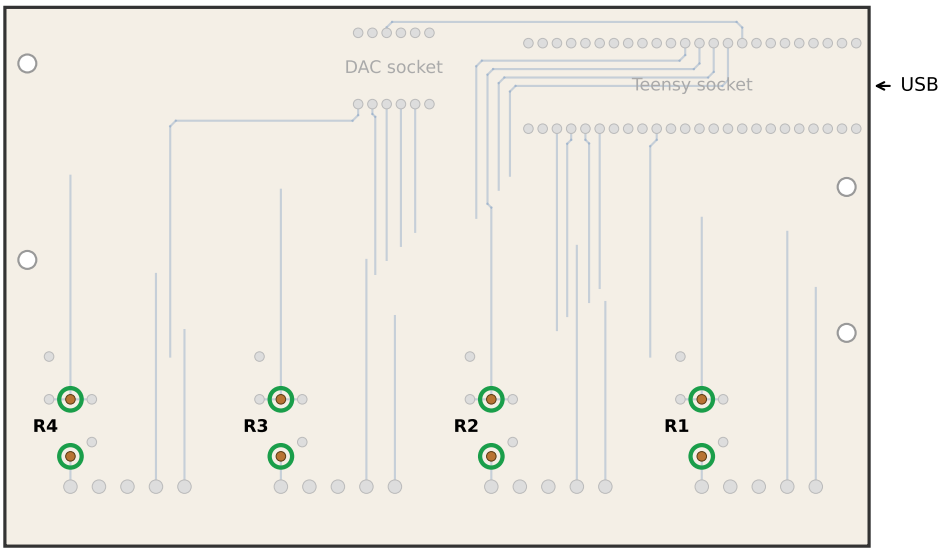
Drawer "1 K": brown-black-black-BROWN-brown. Measure each: 0.99-1.01 kOhm. Either way round. Holes 10 mm apart, one resistor above each module plug (the right-hand of the two resistors there).

TOP (parts side): put the part in here



plug edge: towards you

BOTTOM (turned over, Teensy now on the right): green rings = solder here



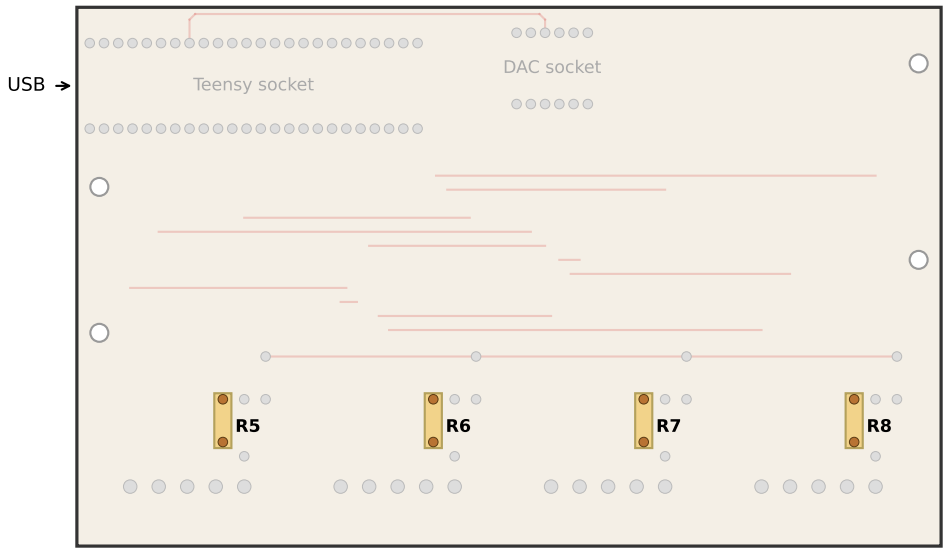
plug edge: towards you

1. Bend both legs down at 90 degrees close to the body so they match the two holes (10 mm apart).
2. Push in from the top until the body lies flat; bend the legs out a little underneath so it stays.
3. Turn over, solder the 8 green-ringed pads (iron on pad and leg 2 s, feed solder, 1-2 s more, iron away).
4. Cut the legs ~1 mm above the joint. KEEP the offcuts: they are the wire for step 4.

Step 2: R5-R8, 1 MOhm (4 resistors, 8 joints)

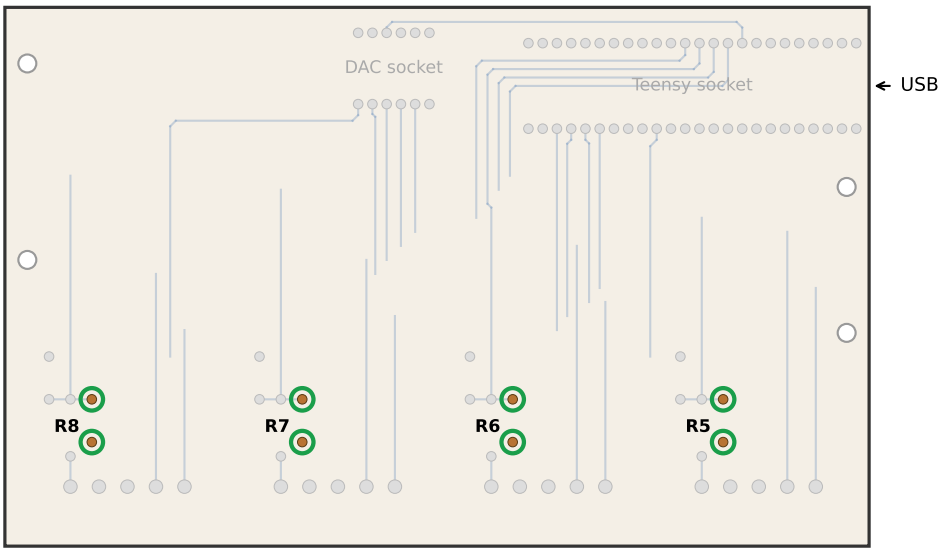
Drawer "1 M": brown-black-black-YELLOW-brown. Measure each: 0.99-1.01 MOhm. Either way round. Holes 7.6 mm apart, left of each 1 k.

TOP (parts side): put the part in here



plug edge: towards you

BOTTOM (turned over, Teensy now on the right): green rings = solder here



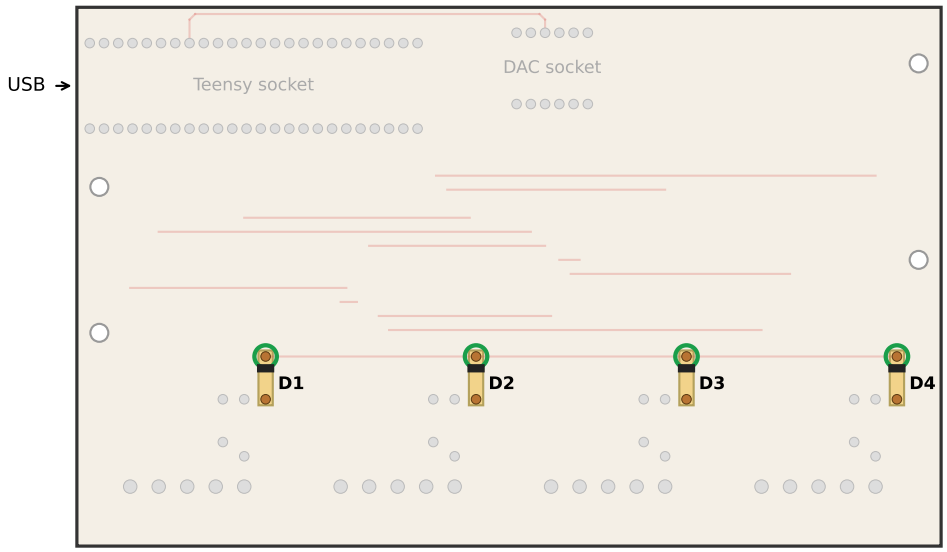
plug edge: towards you

1. Same as step 1: bend, push in flat, turn over, solder the 8 green-ringed pads, cut, keep the offcuts.
2. The lower leg of each is ground (in the copper fill): hold the iron 1-2 s longer there.

Step 3: D1-D4, BAT85 diodes (8 joints + 4 on top)

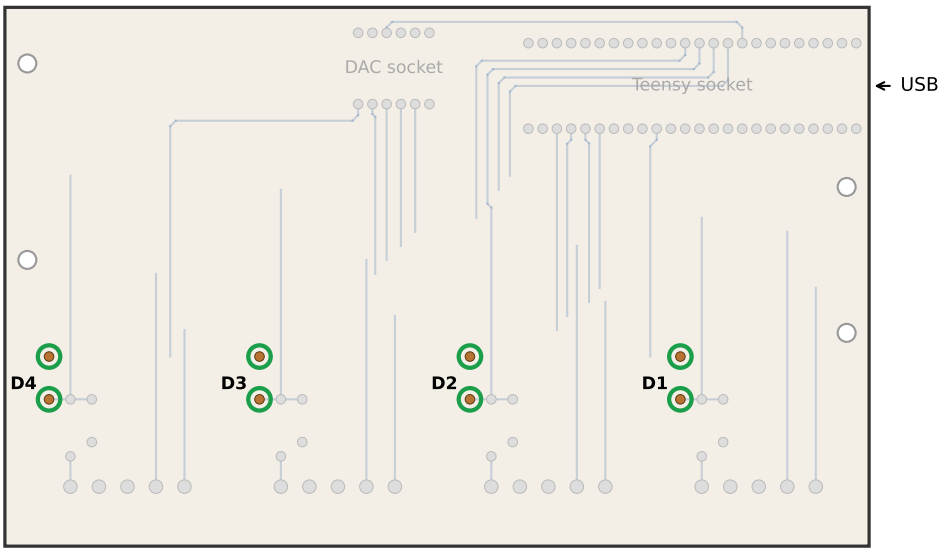
Glass body with a black band. BAND TOWARDS THE FAR EDGE (Teensy side), as the black bar in the picture. Wrong way round shorts the force signal to 3.3 V. Diodes are heat-sensitive: max 3 s per joint.

TOP (parts side): put the part in here; green rings: solder on the top too



plug edge: towards you

BOTTOM (turned over, Teensy now on the right): green rings = solder here



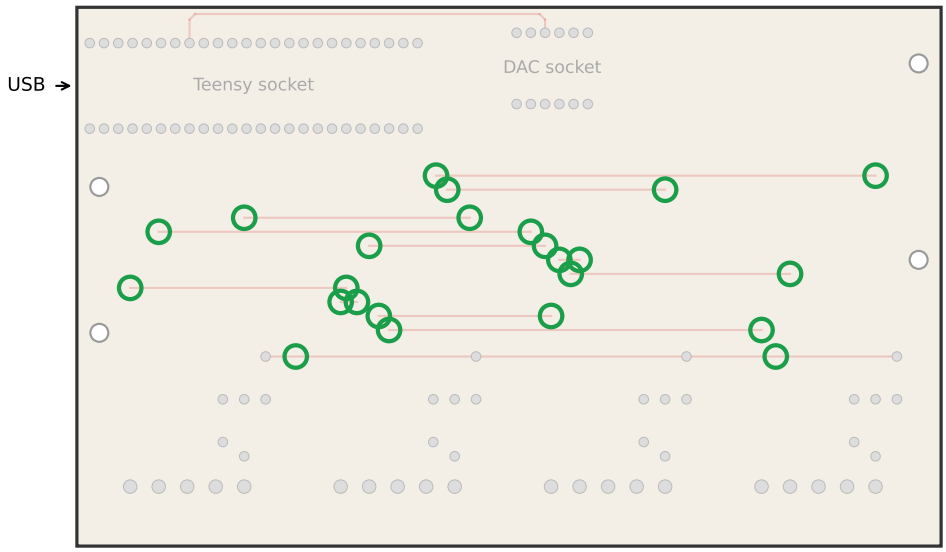
plug edge: towards you

1. Bend the legs gently (not right at the glass: it cracks) to 7.6 mm, band at the upper hole, push in.
2. Turn over, solder the 8 green-ringed pads, cut the legs, keep the offcuts.
3. Turn back to the top: solder the band-end leg ON THE TOP as well (green rings, top picture). The board has no plated holes, and the 3.3 V track that joins the 4 band ends runs on the top.

Step 4: the 24 signal vias (check, wire the missing ones)

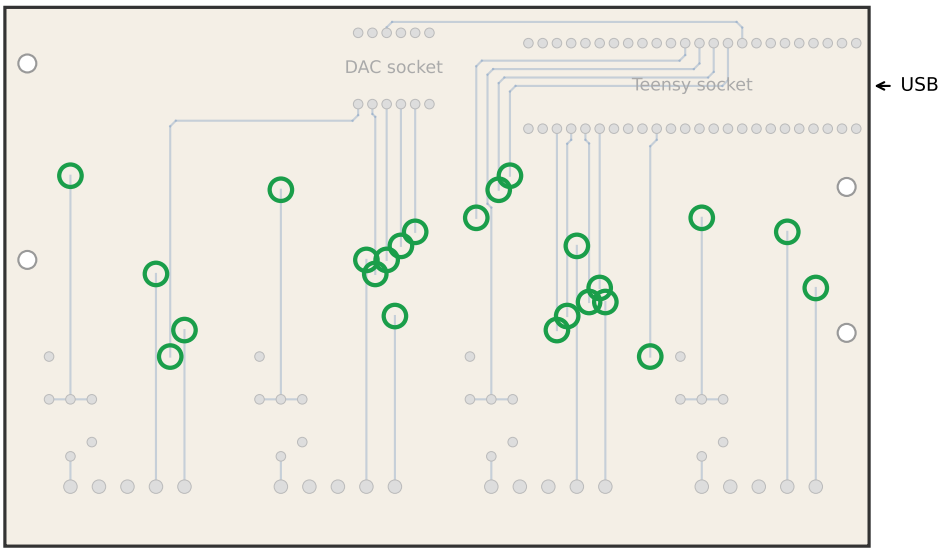
A via is where a track changes side. Yours look done by hand already (small hole with copper in it). Check each one; only the ones that fail need work.

TOP (parts side): put the part in here; green rings: solder on the top too



plug edge: towards you

BOTTOM (turned over, Teensy now on the right): green rings = solder here



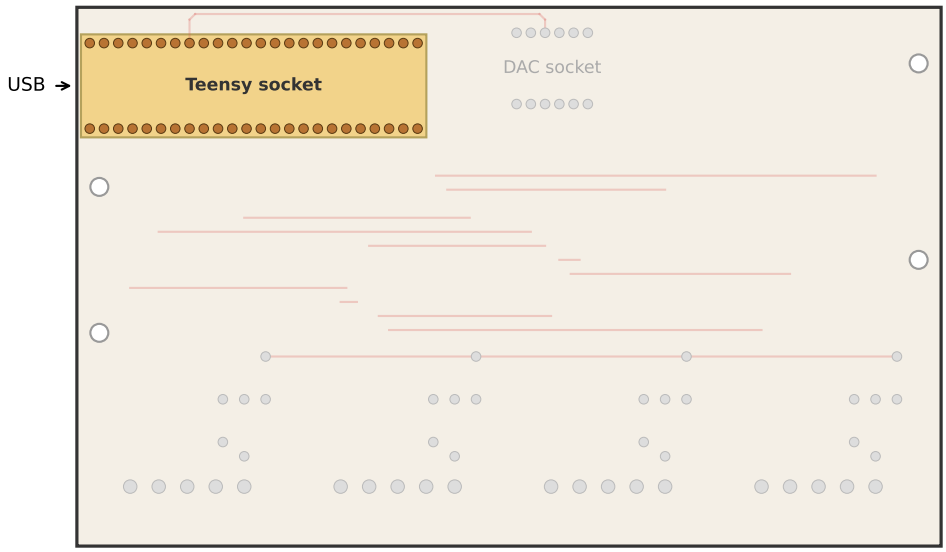
plug edge: towards you

1. Multimeter on beep: one probe on the via's copper on the TOP, the other on its copper on the BOTTOM. Beep = done.
2. No beep: push a step-1/2/3 offcut through the hole, solder it on top and bottom, cut both ends short, check again.
3. The small grey ground vias need nothing.

Step 5: Teensy socket U1 (2 x 24 female headers, 48 joints

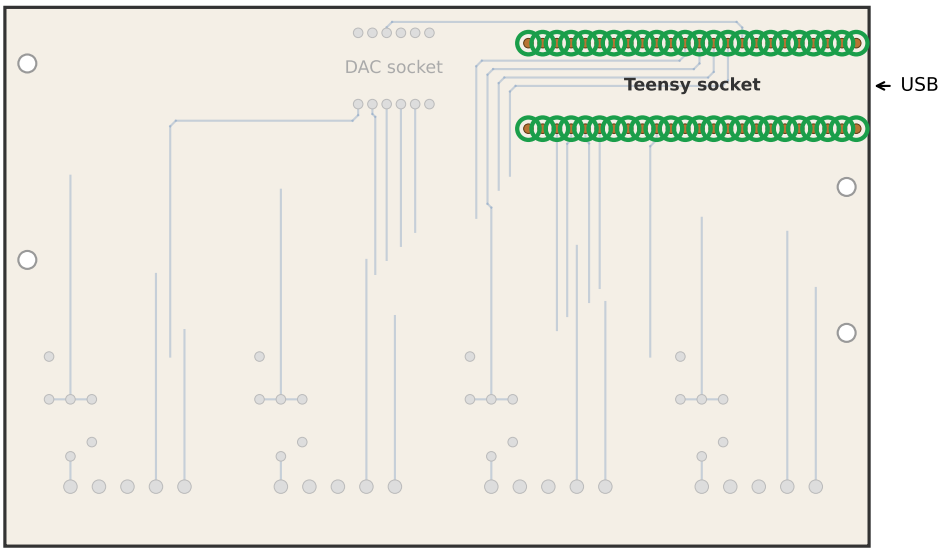
Two 1 x 24 female headers (8.5 mm tall), the long double row at the left. No solder on the top: the one top track here (SCL) is replaced by a wire in step 7.

TOP (parts side): put the part in here



plug edge: towards you

BOTTOM (turned over, Teensy now on the right): green rings = solder here



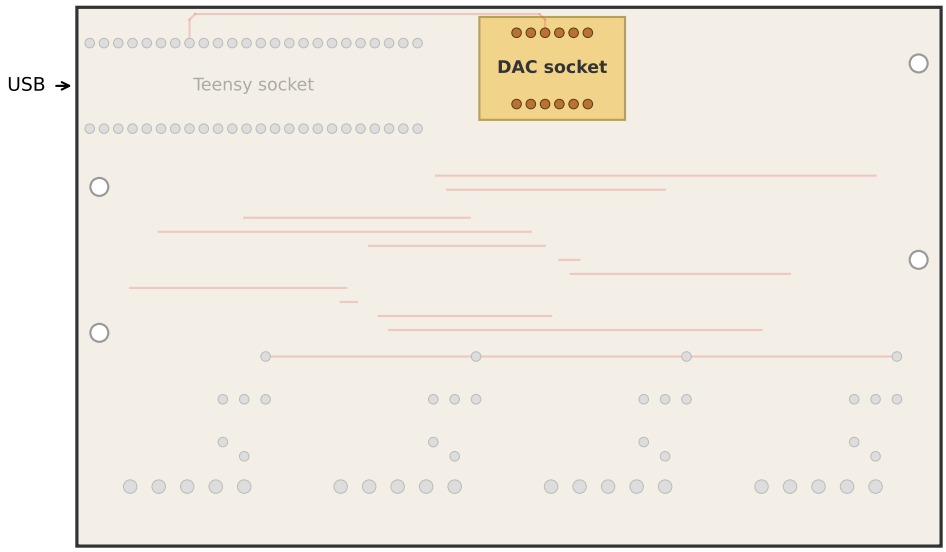
plug edge: towards you

1. Plug both headers onto the Teensy itself (it keeps the rows exactly 15.24 mm apart), then set Teensy + headers into the board from the top, USB at the left edge. Or use spare male strips instead of the Teensy.
2. Turn over (hold the Teensy against the board). Solder ONE pin at each end of each row, check the headers sit flat, re-melt and press if not.
3. Pull the Teensy out (no heat on the Teensy for the rest), then solder all remaining pins. Small joints: pins are 2.54 mm apart; check each row against light for bridges.
4. Solder must not run up into the socket: little solder, iron on the pad, not on the pin top.

Step 6: DAC socket U2 (2 x 6 female headers, 12 joints)

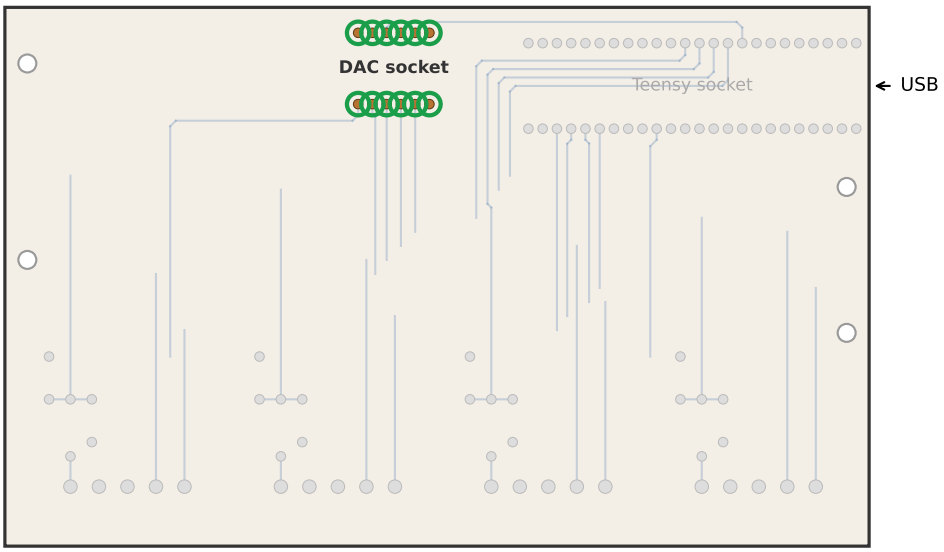
Two 1 x 6 female headers, the short double row near the far edge, right of the Teensy.

TOP (parts side): put the part in here



plug edge: towards you

BOTTOM (turned over, Teensy now on the right): green rings = solder here



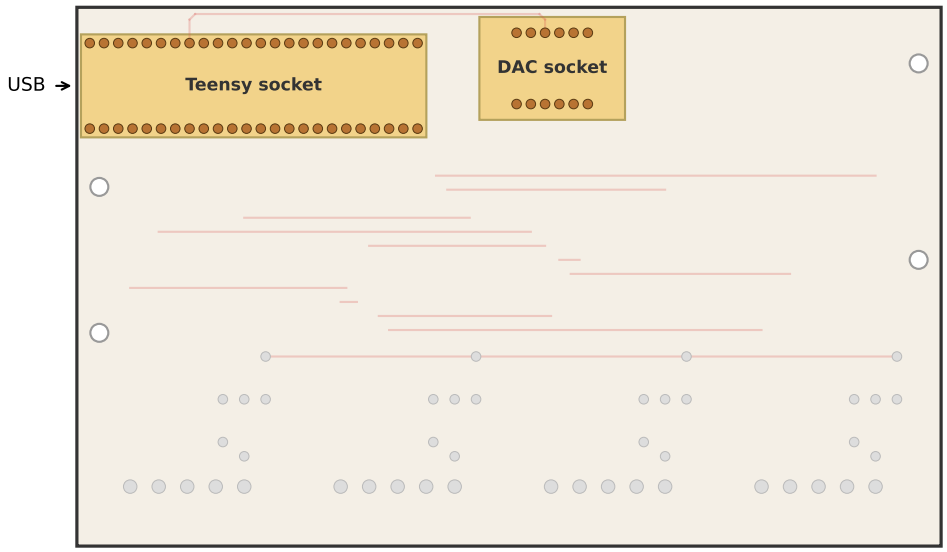
plug edge: towards you

1. Same as step 5: plug them onto the DAC board, set it in, solder one pin per end, check flat, pull the DAC out, solder the rest.

Step 7: SCL wire (1 wire, 2 joints, on the bottom)

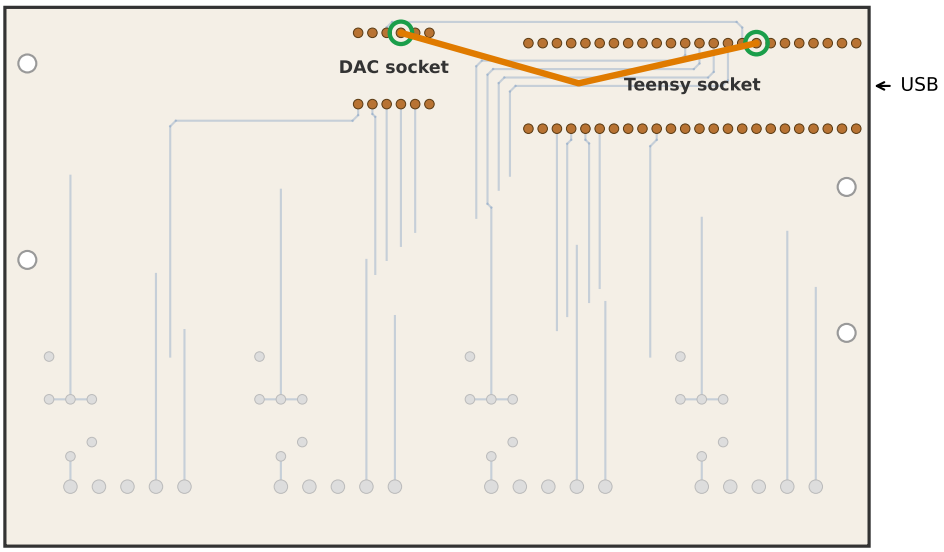
Replaces the top track the sockets cover: Teensy pin 19 (SCL) to the DAC's SCL. Seen from the BOTTOM, plug edge towards you: Teensy socket far row, 8th pin from the RIGHT; DAC socket far row, 3rd pin from the RIGHT.

TOP (parts side): put the part in here



plug edge: towards you

BOTTOM (turned over, Teensy now on the right): green rings = solder here



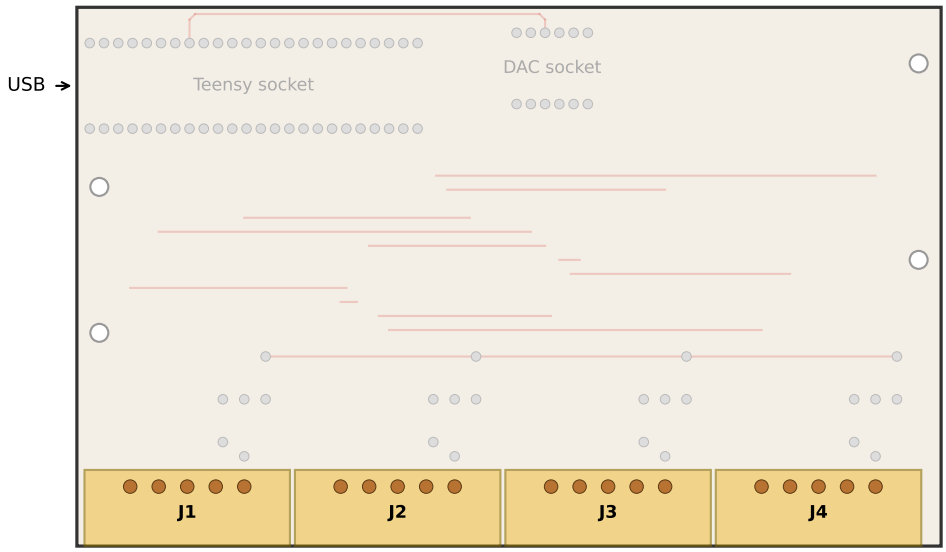
plug edge: towards you

1. Thin insulated wire, about 7 cm. Strip 2 mm at each end, tin it.
2. Lay each end against its pin's existing joint and re-melt the joint around it (orange line in the picture). Route the wire flat, away from other pins.
3. Check: beep from one pin to the other. Then beep each pin to its neighbours: no beep.

Step 8: Module plugs J1-J4 (4 Phoenix MSTB headers, 20 j

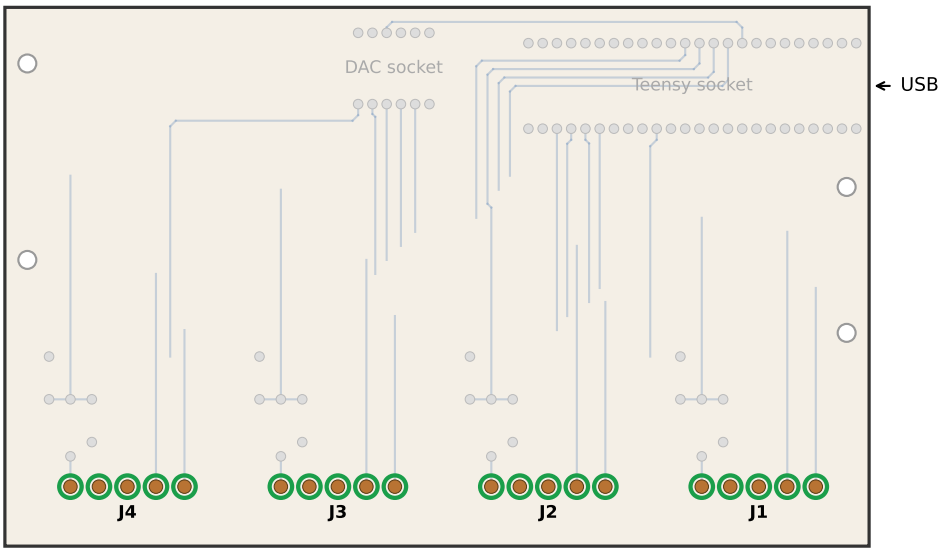
Green 5-pin headers along the near edge, one per module (A-D from the left, seen from the top). The opening where the cable plug goes in faces OUT, over the board edge.

TOP (parts side): put the part in here



plug edge: towards you

BOTTOM (turned over, Teensy now on the right): green rings = solder here

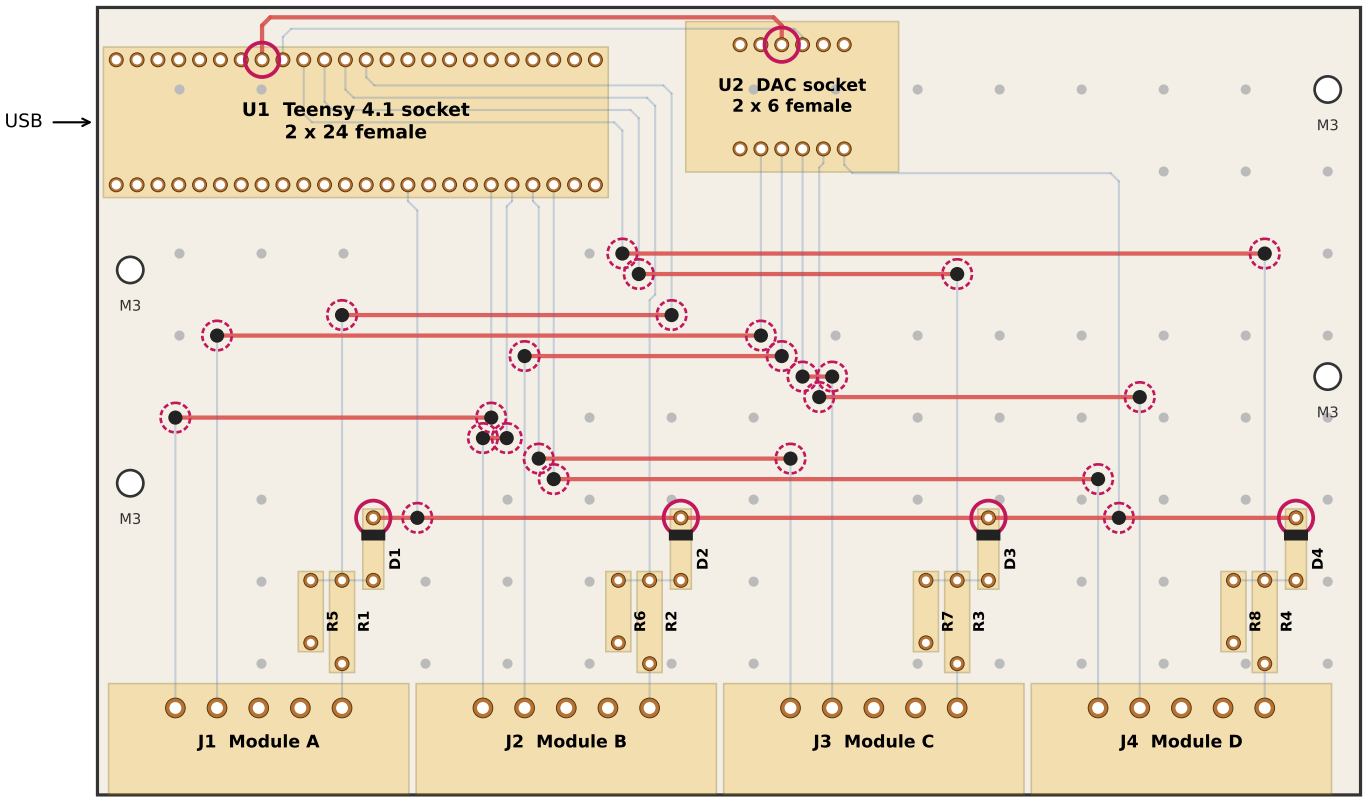


plug edge: towards you

1. Push each header in from the top, opening facing the near edge, flat on the board.
2. Turn over, solder one end pin, check it sits flat and square to the edge, then the other 4. Big pins in big pads: hold the iron 3-4 s, use enough solder for a full cone.
3. Pins 3 and 4 are ground (in the copper fill): they need the longest.

Whole board: TOP side

Seen from above, parts side up, module plugs at the bottom of the picture. Push each part in from this side; its pins come out on the bottom.



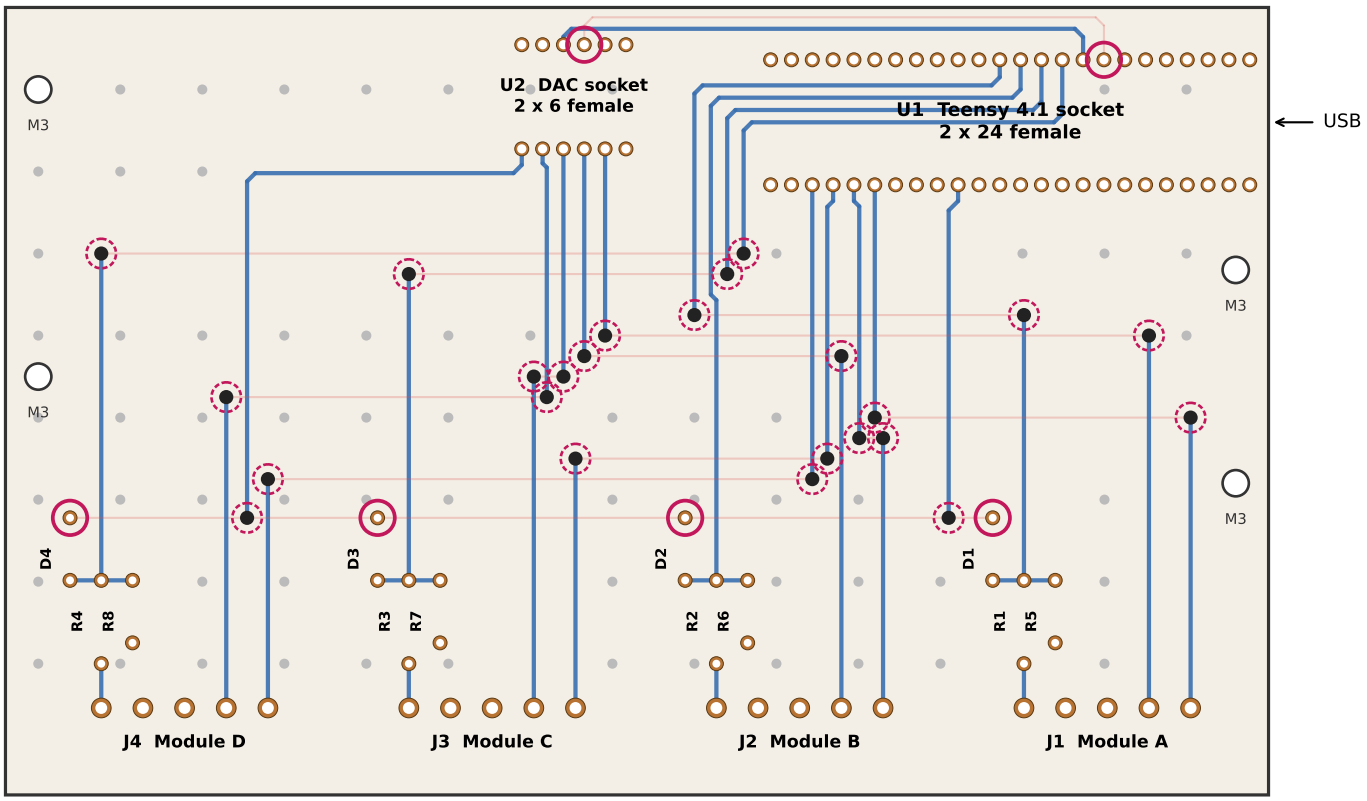
module plugs: this edge (cables plug in here)

where the part body sits
top copper track (on this side)
bottom track (other side, seen through)
pad with hole (one pin each)
via: track changes side here (small grey dots: ground vias, nothing to do)
circled: needs solder ON TOP too, if the holes are not plated (page 13)

1. Resistors and diodes: bend the leads to the hole spacing, push them in until the body lies on the board, bend the leads out a little underneath so they stay, solder, cut.
2. Diodes D1-D4: the black band (black bar in the picture) goes to the hole nearer the far edge (the 3.3 V line). Wrong way round, it shorts the force signal to 3.3 V.
3. Sockets: plug the female headers onto the Teensy (and the DAC) or onto spare male pin strips, set them in, solder one pin at each end, check they sit flat and straight, then solder the rest.
4. MSTB plugs J1-J4: the opening where the cable plug goes in faces the near board edge; the flange screws are at its ends.

Whole board: BOTTOM side

The board turned over (left and right swap: the Teensy is now on the RIGHT). Every pad here gets one solder joint: heat pad and pin together 2-3 s, feed solder until it forms a small cone, cut the leads ~1 mm.



module plugs: this edge (cables plug in here)

- bottom copper track (on this side)
- top track (other side, seen through)
- pad with hole (one pin each)
- via: track changes side here (small grey dots: ground vias, nothing to do)
- circled: needs solder ON TOP too, if the holes are not plated (page 13)

1. Small joints, no bridges: socket pins are only 2.54 mm apart. After each row, look at it against light.
2. Ground pads sit in the copper fill and take longer to heat: hold the iron 1-2 s more.
3. Circled pads and vias: if the holes are not plated, they also need the top-side work on page 13.

Unplated holes: the details

In a plated board the copper inside each hole joins the top and bottom. Without it, a joint on the bottom does not reach a top track. Two places need extra work (circled in pink on pages 11 and 12):

A. 6 pads with a track only on the TOP: solder them on the top as well

- D2 pad 1 (diode, band end): +3V3
- U1 pad 41 (Teensy socket, Teensy pin 19): SCL
- D1 pad 1 (diode, band end): +3V3
- U2 pad 3 (DAC socket, its SCL pin): SCL
- D4 pad 1 (diode, band end): +3V3
- D3 pad 1 (diode, band end): +3V3

Diodes D1-D4: the band-end lead is reachable next to the body: solder it on the top as well as underneath (the top track joins the 4 diodes to 3.3 V). SCL (U1 pad 41 = Teensy pin 19, far row, 8th from the left; U2 pad 3 = DAC far row, 3rd from the left): the top track runs only between these two pins and the socket plastic covers it. Easiest: ignore that track and solder a short insulated wire on the BOTTOM from the one pin to the other.
Ground needs nothing: every ground pin reaches the bottom copper fill, which is one piece (checked on the board file, 2026-10-26).

B. 24 signal vias: a short wire through each, soldered on both sides

A via is where a track goes from one side to the other. Do them right after the resistors and diodes (no via sits under a part), before the sockets: push a cut-off resistor or diode lead (or thin solid wire, max 0.7 mm) through, solder on top and bottom, cut both ends short. The 12 parts give exactly 24 offcuts; keep them. Grey vias (ground): leave them.

- +3V3: (125, 62) mm, (39, 62) mm
- EN_A: (48, 50) mm, (10, 50) mm
- EN_B: (50, 52) mm, (47, 52) mm
- EN_C: (54, 55) mm, (84, 55) mm
- EN_D: (122, 58) mm, (56, 58) mm
- F_A: (30, 38) mm, (70, 38) mm
- F_C: (66, 32) mm, (105, 32) mm
- F_D: (142, 30) mm, (64, 30) mm
- SP_A: (15, 40) mm, (81, 40) mm
- SP_B: (52, 42) mm, (83, 42) mm
- SP_C: (90, 45) mm, (86, 45) mm
- SP_D: (127, 48) mm, (88, 48) mm

Positions in mm from the top-left corner seen from the top (USB edge left, plug edge at the bottom). Check each with the multimeter afterwards: beep from the pad on one side to the track it should reach on the other.

Step 9: check before plugging anything in

Multimeter on beep. No Teensy, no DAC, no modules (README 'Assembly and first power-up').

1. 3V3 to GND must NOT beep: Teensy socket 3V3 pin to any J pin 3 or 4.
2. Each plug J1-J4: pins 3 and 4 to a Teensy socket GND pin must beep.
3. J1 pin 1 to Teensy socket pin 27 must beep; J2 pin 1 to 28, J3 to 29, J4 to 30 (module enable).
4. DAC socket SCL to Teensy pin 19 and SDA to pin 18 must beep (these use the top track and a via: steps 4 and 7).
5. Each J pin 2 (setpoint) to its DAC output pin must beep (A-D: through vias).
6. Then plug in the DAC and the Teensy (USB at the board edge), USB only: bumpem info --port <PORT> must answer.
7. One module at a time, 48 V off: plug in, then scripts/open_loop_test.sh for that module.